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THEORETICAL NOTE

Exploring the Underlying Psychological Constructs of Self-Report Eating Behavior Measurements: Toward a Comprehensive Framework

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Food and eating are fundamental for survival but also have significant impacts on health, psychology, sociology, and economics. Understanding what motivates people to eat can provide insights into "adaptive" eating behavior, which is especially important due to the increasing prevalence of health-related conditions such as obesity. There has been considerable interest in developing theoretical models and associated constructs that explain individual differences in eating behavior. However, many of these models contain overlapping theories and shared theoretical mechanisms of action. Currently, there is no recognized standard framework that integrates psychological, physiological, and neurobiological theory to help explain human eating behavior. The aim of the current article was to review key psychological theories in relation to energy balance, homeostasis, energy intake, and motivation to eat and begin to develop a comprehensive framework of relevant factors that drive eating behavior. The key findings from this review suggest that eating behavior is conceptualized by elements of dual process models, which include conscious processing (reflective factors) and automatic responses to desires, environmental cues, habits, and associative learning. These processes are mediated by neurobiology and physiological signaling (homeostatic feedback) of energy balance, which is more tolerant of positive than negative energy balances. From a synthesis of available evidence, it is suggested that eating behavior constructs (traits) can be explained by three latent constructs: reflective, reactive, and homeostatic eating. By understanding the interplay between reflective, reactive, and homeostatic processes, interventions can be developed that tailor treatments to target key aspects of eating behavior.

Keywords: eating behavior, obesity, reactive eating, reflective eating, homeostatic eating

Eating behavior encompasses a complex array of physiological, psychological, environmental, and cultural factors. Food and eating are central to everyday life (Warde, 2016), whereby their meaning and usage extend far beyond nutritional maintenance (Rozin, 1999). Food is a social vehicle; it involves symbolic functions; it takes on moral significance; and it is a medium for aesthetic expression (Rozin, 2005). Shifts in the food system have led to growing interest in food, eating, and body weight across the globe, partly due to the increasing prevalence of health-related diseases that are attributable to foods and diets (Warde, 2016). Part of the complexity in understanding human eating behavior involves the psychology of why we eat. Thus, understanding what motivates people to eat can help to shed light on what adaptive eating behavior involves, as well as what motivates people to undereat or overeat.

Researchers have developed numerous theoretical models and generated a panoply of constructs that explain the factors that motivate eating behavior and facilitate overeating. These "eating behavior traits" (EBTs), quantified by psychometric, self-report instruments, relate aspects of eating behavior to overeating and obesity (French et al., 2012) and are considered to be reliable indices of food-related behaviors (Llewellyn & Wardle, 2015). EBTs can also be used to phenotype individuals based on their endorsement of specific eating behavior motivations, allowing for the examination of the moderators and mechanisms that contribute to overeating and obesity. However, several theories of eating behavior share similar constructs and proposed mechanisms of action. This is a source of confusion since the psychometric and predictive validity of specific EBTs are unknown. Further, researchers use different measures for the same purposes as well as the same measures for different purposes, which limits comparability of studies, meta-analyses, and scientific insight. For example, questionnaires that measure emotional eating are used to assess both emotional eating and have

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been used as proxy measures of food intake (Devonport et al., 2019). This use of different questionnaires that measure the same constructs is termed a jangle fallacy (Kelley, 1927), and previously, it has been suggested that jangle fallacy is common in obesity research (Vainik et al., 2015). The use of varying questionnaires also makes it challenging to compare studies that use different questionnaires for the same or similar EBTs. These numerous scales could be adding unnecessary heterogeneity in cause–effect relationships and adding unnecessary burden on participants who are required to fill in multiple, time-consuming measures.

Currently, there is no agreed framework or standardized set of measures that describe motivations for eating. Thus, the aim of this conceptual article is to draw from key psychological, physiological, and neurobiological models as a starting point to develop a comprehensive framework that identifies the core, latent constructs underlying EBTs. In a previous systematic review and meta-analysis conducted by our research team, we identified several constructs that measure EBTs and tested the extent to which these EBTs were associated with body mass index (BMI) and energy intake (C. Dakin, Beaulieu, et al., 2023). Having examined these EBTs, we identified that not all the concepts related to motivations for eating are covered in many of the current EBT measures. Therefore, the purpose of this review is to propose a provisional conceptual framework and examine it in the context of existing psychological theories, which covers some of the areas missed by the previous systematic review. It is important to clarify the theories that relate to each EBT and the relationships between theories and EBT constructs. Thus, while the field of EBTs may not need any new theories, there is a need to better understand the measures that currently exist, how they relate to each other, how they relate to physiology and neurobiology, and, as trait measures, longer term indices of energy balance. An assumption we make is that most longer term changes in energy balance are driven by behavior (R. J. Stubbs et al., 2023). This framework examines the underlying common constructs that are shared between numerous overlapping EBT measurements and attempts to align existing theory with existing EBT constructs. This approach of aligning these empirically derived constructs with existing theory allows for clarification and standardization of measurements that will enable better comparisons of studies, analyses, and effect sizes that cannot currently be determined because the field is overwhelmed with overlapping measurements (Michie, 2014). The measurement of physiological variables and the integration of objective tracking of eating behavior with self-report measures of eating behavior are important (R. J. Stubbs et al., 2021). However, the current article is constrained to a discussion of self-report measures of EBTs. This does not diminish the importance of other types of measures, but these are not the focus of the current article.

This article begins with some key definitions of human appetite and eating behavior to establish the discussion parameters (see Human Appetite and Eating Behavior section). The next sections summarize and evaluate key psychological theories (see Psychological Accounts of Eating Behavior section), physiological (see Physiology of Eating Behavior section), and neurobiological models that have been proposed to explain motivations for eating (see Neurobiology of Eating Behavior section). Last, the constructs from these scientific domains will be synthesized to propose a framework that can be used to describe and quantify human EBT measurements (see Eating Behavior Traits Framework section).

Human Appetite and Eating Behavior

Human appetite is often used rather loosely to describe quantitative and qualitative aspects of motivation to eat and ingestive behavior. In its broadest definition, appetite covers the whole field of food intake, food selection, motivation, and preference. This includes qualitative aspects of eating, sensory aspects, or responsiveness to environmental stimulation and eating in response to physiological stimuli and energy deficit. It is also common to see appetite subdivided into "homeostatic"—which is regulated and links the physiological needs for energy and nutrients with the behavior that satisfies these needs (eating), shaped by physiological *excitatory* and *inhibitory* signals—and "hedonic"—which is reward-driven signals and links pleasant thoughts and cravings about food and sensory appreciation of certain food attributes with the expression of food preference and choice. Hedonic appetite reflects both liking and wanting for food. Eating is a measurable form of motivated behavior, and motivations to eat can be biologically driven as well as environmentally influenced, for example, social motives to eat (Watts et al., 2022). The psychobiology of motivation to eat, as with other goal-oriented motivated behaviors, follows cycles of anticipation, consummation, and cessation, which can be influenced by various mechanisms, described as follows:

Hunger (Motivation to Eat)

Hunger is defined as the experienced subjective motivation to consume, which includes provoking and sustaining a behavioral response to eating, often believed to be in response to a biological need. Hunger functions as an intervening variable in the stimulus–response sequence that is responsible for initiating eating (Watts et al., 2022). While hunger initiates eating, there is no single mechanism of hunger.

Satiation (Termination of Eating)

The processes that occur during a meal that generate negative feedback that leads to the termination of a meal are termed satiation (within-meal inhibition).

Satiety (Lack of Motivation to Eat)

Satiety is defined as between-meal inhibition. Satiety further suppresses the drive to consume and determines the length of the intermeal interval (suppresses postmeal intake).

Energy Balance

Energy balance is the difference between energy intake and energy expended and excreted over a given period of time (Blaxter, 1989). Therefore, energy storage is equal to intake minus expenditure:

$$\begin{aligned} \text{Energy Intake} - \text{Energy of Faeces} - \text{Energy of Urine} \\ - \text{Energy of Combustible Gas} \\ = \text{Energy Produced.} \end{aligned} \tag{1}$$

This equation is frequently simplified to give:

$$\Delta\text{Energy Storage} = \Delta\text{Energy Intake(EI)} - \Delta\text{Energy Expenditure(EE)}. \quad (2)$$

Dietary macronutrients summate to determine energy intake and influence energy balance regulation in humans through physiological effects and voluntary food intake (R. J. Stubbs et al., 2023). Macronutrients can act as powerful unconditioned stimuli, modifying feeding responses and forming learned food preferences related to pleasure (Berridge & Kringelbach, 2008; Kringelbach et al., 2012). Energy expenditure (total daily energy expenditure) is the heat released by the body through resting metabolism (resting metabolic rate [RMR]), the thermic effect of food, physical activity (PA), and nonexercise activity energy expenditure (Kleiber, 1961). Energy storage is the potential chemical energy that is mostly stored as fat but can also be stored as glycogen, protein, and heat energy due to body temperature changes (Kenny et al., 2008). The pathways to and from obesity appear to be largely behavioral, involving multiple mechanisms (R. J. Stubbs et al., 2023). Indeed, there is consensus that increased population-level obesity development is primarily caused by excessive energy intake (Prentice et al., 1989). This means that the key intervention targets by which eating behavior and energy balance can be altered appear to be largely related to energy intake rather than energy expenditure (R. J. Stubbs et al., 2023). As such, several theories have been developed to explain how several different factors of the act as mechanisms to influence food and energy intake (Blundell & Stubbs, 1999; Blundell & Tremblay, 1995; Prentice et al., 1989; R. J. Stubbs, 1998).

EBTs

EBTs are constructs that have been theoretically developed to explain a defined eating style or disposition to eat. They are measured using self-report, psychometric questionnaires such as the Three-Factor Eating Questionnaire, which measures three EBTs: dietary restraint, eating disinhibition, and susceptibility to hunger (A. J. Stunkard & Messick, 1985). While they are not direct measures of eating behavior, EBTs are considered to be reliable indices of individual differences in motivation to eat or not eat for different reasons (Llewellyn & Wardle, 2015). For example, the child eating behavior questionnaire, which measures several EBTs, including responsiveness and satiety responsiveness, has been validated using behavioral measures (Carnell & Wardle, 2007) and has been shown to be stable over time (Ashcroft et al., 2008). Llewellyn and Wardle (2015) have also shown that EBTs play a causal role in excess weight gain. For example, a population-based pediatric birth cohort of twins has been used to prospectively test the hypothesis that appetite plays a casual role in the development of obesity. The results found that high food responsiveness and low satiety responsiveness predispose children to obesity. The use of longitudinal data made it possible to test the causal direction between appetite and adiposity. Furthermore, EBTs have also been used to characterize eating behavior in relation to disordered eating patterns and predisposition to obesity (French et al., 2012).

Psychological Accounts of Eating Behavior

In this section, key psychological theories are discussed in relation to eating behavior. Their central elements, key constructs, and mechanisms of action are presented in Table 1.

Psychosomatic Theory

H. I. Kaplan and Kaplan (1957) argued that the development of obesity is attributed to overeating, caused by emotional disturbances that serve to increase food intake, and this is termed a psychosomatic disorder. The underlying assumptions of psychosomatic theory originate from learning theory (Dollard & Miller, 1950; Mowrer, 1950), whereby overeating is viewed as a learned behavior that is acquired as a strategy for reducing anxiety. The authors proposed two types of abnormal overeating: one that is characterized by feelings of hunger and another that is not associated with excessive hunger and is instead attributed to avoidance of anxiety (H. I. Kaplan & Kaplan, 1957). A normal response to negative emotions such as anxiety, fear, and anger is a loss of appetite (Cannon, 1953). However, some individuals respond to these emotions by excessively overeating.

Compulsive overeating can be understood from the perspective of drive reduction because eating can reduce a drive (e.g., hunger) and is rewarded and therefore learned. Emotions can also become drive states, meaning that eating can be associated with a reduction of negative emotions such as anxiety, and thus the act of eating is rewarded. This means an individual can learn to eat in response to hunger and anxiety, which could lead to overconsumption and obesity. Bruch (1964) further suggested that hunger cues can be confused with emotional states, which can result in excessive overeating. She also highlighted the importance of early learning experiences that play a role in associating hunger with a pattern of cues that lead to eating behavior. For example, if an individual comes from a home where their mother uses food as a means of control, it may increase their chance of using food to reduce anxiety (Bruch & Touraine, 1940).

While H. I. Kaplan and Kaplan (1957) argued that overeating reduces negative emotions and there is evidence that individuals may use food as a means of reducing anxiety/stress, for example (Bruch & Touraine, 1940), there are limitations to this theory. One issue with the anxiety-reduction explanation of obesity is that if overeating is driven by an aversive state and also acts to reduce an aversive state, it is unclear what is maintaining and reinforcing the behavior. Robbins and Fray (1980) argued that overeating in individuals with obesity cannot be maintained by anxiety reduction, and instead, the overeating behavior acquires its own positive reinforcing properties, termed food reward. Furthermore, psychosomatic explanations cannot account for evidence that some individuals reduce their eating when experiencing high levels of stress. The authors conclude that overeating can be provoked by stress but that eating itself does not act to reduce stress, which disagrees with psychosomatic theory. Overall, the idea that an individual can learn to overeat as a means of anxiety reduction can explain the eating behavior of some individuals with obesity. However, there are flaws and limited generalizability with this explanation of obesity, which has led to other theories emerging in the literature.

Table 1
Summary of Key Constructs and Theories of Eating Behaviour

Eating behavior theory	Central element to theory	Main construct	Mechanism of action
Psychosomatic theory	Overeating is viewed as a learned behavior that is used as a means of reducing anxiety.	Emotional eating	Emotions become drive states, meaning that eating is associated with a reduction of emotions such as anxiety.
Externality theory	Eating behavior of individuals with obesity is defined by increased responsiveness to external cues and decreased responsiveness to internal/physiological signals.	Internal control External control	Responsivity to environmental prompts and cues.
Set point theory	Individuals with obesity are in a state of chronic energy deficit because they are attempting to keep their body weight below their biological set point.	None	Hypothalamic centers defend different baselines in different individuals.
Restrained eating	Individuals with high levels of restraint behave externally once restraint has been discarded, and this triggers further eating. The degree of deprivation (restraint) predicts eating behavior.	Dietary restraint	Conflict between cognitive control and "homeostatic" signaling.
Hedonic hunger	Hedonic hunger is the desire to consume palatable foods in the absence of physiological hunger.	Hedonic hunger	Hedonic hunger acts to increase storage of body fat, which protects the body from food scarcity.
Dual process theory	Social behavior is controlled by reflective (cognitive reasoning) and impulsive systems (fast, inflexible, no attentional resources required).	Impulsive Reflective	Reflective behavior is the consequence of a decision process. Impulsive behavior is elicited through spreading activation.
Dual pathway model	Restrained eating and negative affect compete to control eating behavior.	Restraint Negative affect	A lack of interoceptive awareness means individuals confuse distress with hunger.
Goal conflict model	Restrained eaters cannot regulate their food intake due to conflict between two incompatible goals.	Eating enjoyment goal Weight control goal	Palatable food cues prime the eating enjoyment goal. Deliberate eating behavior, for example, dieting primes the weight control goal.
Behavioral susceptibility theory	Increased food responsiveness and decreased satiety responsiveness leads to overeating.	Food responsiveness Satiety responsiveness	Genetic risk operates through these FR and SR, and this results in differences in susceptibility to the food environment.
COM-Bmodel	A framework of behavior change that involves three essential conditions.	Capability Opportunity Motivation	A range of mechanisms are involved in this framework, including the internal and external environments.

Note. This table summarizes the key theoretical constructs and where relevant psychological theories that can be applied to eating behavior, including the central elements, main constructs, and mechanisms of action of each theory. FR = food responsiveness; SR = satiety responsiveness; COM-B = capability, opportunity, motivation–behavior.

Externality Theory

Externality theory suggests that eating behavior in individuals with obesity is influenced by external cues (which are unrelated to nutritional need states) and less influenced by internal cues. In contrast, normal-weight individuals are influenced by both internal and external cues (Schachter, 1967). Responsiveness to external cues was proposed as a direct function of weight, with responsiveness to internal cues as an inverse function of weight (Nisbett, 1968b). Evidence supports the internal hypothesis, showing weaker correlations between gastric mobility and hunger in individuals with obesity (A. Stunkard & Koch, 1964). Furthermore, individuals with obesity did not increase food intake after deprivation, while normal-weight individuals did (Schachter et al., 1968), suggesting that they are less able to respond to their internal/physiological cues of hunger. However, some studies contradict this theory, finding no difference in internal regulation between weight groups (Wooley, 1972) and poor regulation of food intake in normal-weight individuals as well (Spiegel, 1973).

Nisbett (1968a) tested the external hypothesis and found that participants with obesity ate more when presented with more food, while normal and underweight participants did not change their intake depending on the amount of food presented. Similarly, manipulating a clock time determined the amount eaten in individuals with obesity but had no effect on normal-weight participants (Schachter & Gross, 1968). This suggests that individuals with obesity are more susceptible to external cues and immediate food stimuli. Externality theory has been criticized for being too simplistic (Rodin, 1981). For example, externality can be found in individuals in all weight categories, which can lead to overeating, but only in specific conditions. Indeed, Meyers and Stunkard (1980) found no evidence to support the hypothesis that individuals with overweight or obesity are more responsive to food cues than individuals who do not live with overweight or obesity. Additionally, internal sensitivity is not a unique characteristic of individuals within a normal weight category. This means that the degree of weight gain and obesity is dependent on several factors, not just externality. Rodin (1981) also suggested that the simplistic internal–external dichotomy is not empirically supported. Additionally, the key studies in support of externality theory used relatively small sample sizes and short study durations (one test day). This has led to the development of other theories that address its limitations.

Set Point Theory

Extending from externality theory, Nisbett (1972) argued that some individuals with obesity are below their biological obese "set-point" and are hungry because they are attempting to keep their weight below this set point. From a physiological perspective, the set point model mainly focuses on the importance of fat mass for the feedback loop. This is supported by the discovery of leptin and the associated pathways that provide the link between adipose tissue and the central nervous system (Campfield et al., 1996). Set point theory is based on the belief that hypothalamic centers defend adipose tissue set points, and these set points have different baselines in each individual. This suggests that an individual could be living with obesity but be below their biological set point, which the central nervous system is trying to defend. If an individual without obesity is below their set point, their eating behavior should reflect that of an individual with obesity, assuming the person with obesity is below their own "very-obese set point." However, if an individual with obesity is at their set point, their eating behavior should reflect normal eating. Nisbett (1972) suggested that an individual with obesity who is below their set point is in a state of deprivation, and this is what generates increased responsiveness to external cues.

Set point theory gained attention from researchers; however, since the 1970s, the proportion of the population that is obese has risen dramatically, and this rise is projected to continue (Agha & Agha, 2017). Applying set point theory, this would indicate that a significant proportion of the population has a set point in the obese range. Set point theory does not account for why there is an increasing number of individuals with an "obese set-point." Furthermore, there is a lack of evidence to support the main components of this theory and limited capacity for the theory to identify an individual's purported set point. In a study testing responsiveness to external food cues before and after weight loss through dieting, Rodin et al. (1977) found that across three experiments, weight loss was unrelated to changes in responsiveness to external cues. Experiment 1 included 85 overweight females who completed 8 weeks of daily exercise and restriction of food intake. The intervention led to a mean weight loss of 12.55 kg in participants with obesity, 8.82 kg in participants with overweight, and 2.55 kg in normal-weight participants. The results showed that externality scores change very little before and after weight loss and that any changes that did occur were not related to the amount of weight lost. Overall, in contrast to Nisbett's hypothesis, the authors conclude that external responsiveness is not a function of the degree of obesity.

More recently, other models have been presented to address the limitations of a strict set point model. The settling point model assumes there is no active metabolic regulation of body weight but behaves as if body weight is being regulated due to passive regulation of body weight "at a point defined by the level of the unregulated parameter (either inflow or outflow)" (Speakman et al., 2011, p. 736). The thrifty gene hypothesis suggests that there is an adaptative usefulness of increased energy storage (Neel, 1962; Wendorf & Goldfine, 1991). Furthermore, the dual intervention point model (Speakman, 2007) suggests there are upper and lower boundaries that distinguish the points where active physiological regulation occurs. Between these points, there is weak or no physiological regulation of weight. Last, the general model of intake regulation (de Castro & Plunkett, 2002) proposes that intake is influenced by uncompensated factors that are not influenced by intake and compensated factors that are. This model does not assume there are any set points for body weight or intake and instead assumes that the level of any factor that is defended is flexible. A change in one or more factors can produce a new defended level. These models also assume an asymmetric defense of the body weight set point, such that negative energy balance is defended more aggressively than positive energy balance (Levitsky, 2005).

Restrained Eating

Herman and Mack (1975) expanded Nisbett's set point theory to explain the eating behavior of normal-weight individuals. They proposed that many normal-weight individuals are "biologically underweight" (below their biological set point), meaning they should overeat because they are below their set point. However, societal pressures lead them to restrain their eating. Restraint theory suggests that individuals who differ in their concern about weight will react differently to restraint manipulation. In an initial experiment, participants with high levels of restrained eating (measured using the restraint scale) consumed more ice cream after a milkshake preload, while those with low levels of restrained eating showed internal regulation, consuming less ice cream with a larger preload. This study indicates that highly restrained individuals exhibit external eating behavior when restrained eating is discarded. The results suggest that it is high levels of restrained eating that lead to external eating, meaning individuals with and without obesity can be externally responsive. Therefore, eating patterns like restrained eating may be better predictors of eating behavior than body weight.

Herman and Mack's restraint theory sparked interest in using self-report measures of eating patterns (EBT) as a way of explaining and predicting eating behavior. While the concept of restrained eating does have supporting evidence, this construct is only a descriptive term and does not provide a mechanism for eating behavior. Overall, the predictions of restraint theory are not fully supported by the literature. For example, restraint theory suggests that engaging in restrained eating stimulates counterregulatory responses, which can cause disinhibited eating (overeating in response to different stimuli) and binge eating behavior (Herman & Mack, 1975; Herman & Polivy, 1980). Restraint theory proposes that individuals who engage in restrained eating may feel deprived (perceived deprivation), and this could make them vulnerable to overconsumption and weight gain. But it does not elaborate on the nature of the deprivation, for example, physiological deprivation, deprivation of rewards, or failure to fulfill other psychological needs. However, other evidence suggests this effect is in the opposite direction, meaning that high levels of restrained eating are not a cause of overeating. Instead, it is the observed consequence of unsuccessful weight management attempts (Schaumberg et al., 2016). There is also evidence against the proposal that restrained eating is associated with overeating and obesity. For example, engaging in a sustained effort to control and monitor food intake (restrained eating) is associated with successful weight maintenance (Johnson et al., 2012). It has since been suggested that inconsistent restrained eating (periods of successful restrained eating followed by disinhibition) can predict the onset of eating disorders and overeating. However, restrained eating that is characterized by consistent and sustainable energy restriction is associated with weight loss and weight maintenance without eating disorder risk (M. T. McGuire et al., 1999; Phelan et al., 2009).

More recently, Polivy et al. (2020) highlighted that defining restrained eating as a construct is a complicated endeavor because research suggests that different questionnaires used to identify restrained eaters may be assessing different constructs (Williamson et al., 2007). The initial restrained eating studies attempted to determine how chronic dieters behaved in various challenging situations. The original concept of restrained eating, as measured by the restraint scale (Herman & Polivy, 1980), was proposed as a way of identifying individual differences in the motivation to cognitively control body weight, with initial suppression of food intake that then succumbs to temptation, overeating, and emotional responses. This means restrained eating was defined as a cycle of restriction and indulgence as a reaction to the demands of chronic dieting (Polivy & Herman, 1985). Consequently, restrained eating was identified as a construct (EBT).

Since the development of the restraint scale (Herman & Mack, 1975), other questionnaires have been developed to measure restrained eating, such as dietary restraint as measured by the Three Factor Eating Questionnaire (TFEQ; A. J. Stunkard & Messick, 1985) and restrained eating as measured by the Dutch Eating Behavior Questionnaire (DEBQ; Van Strien et al., 1986). However, these questionnaires use somewhat different defining measures. TFEQ restraint primarily aims to identify successful dieters who can control their intake (restraint is weighted positively), and DEBQ restraint identifies attempts to control intake and the types of cues that hinder restraint (Polivy et al., 2020, 2023). These scales measure successful restriction over time without measuring the other dimensions (e.g., disinhibition when inhibition fails) that the restraint scale measures that were seen as critical to restraint theory.

Polivy et al. (2023) suggested that because the TFEQ and DEBQ remove disinhibition, emotional, or externally driven eating out of the measurement of restrained eating, they are measuring a different construct from that of restrained eating as measured by the restraint scale. Consequently, these questionnaires may be measuring a construct that could be better termed as restrictive eating (Polivy et al., 2020). In support of this, research has found that the restraint scale is better at identifying individuals who struggle to control their food intake (Adams et al., 2019), and only the restraint scale can consistently identify dieters who overeat in response to emotions (Polivy et al., 2020). In contrast, the TFEQ is more consistently able to predict dietary success over time (Keller & Hartmann, 2016). The difficulty in defining the construct of restrained eating suggests that while dieters share the same goal of losing or maintaining a lower body weight, they differ in personality, motivation, ability, behaviors, attitudes, and self-image.

This conceptual ambiguity needs to be considered when researchers decide which questionnaire to use. It appears that the TFEQ or DEBQ is more appropriate for understanding successful caloric restriction. Whereas assessing cycles of restriction followed by disinhibited behavior and overconsumption is better predicted by the restraint scale and Eating Disorder Examination Questionnaire (EDE-Q) (Fairburn & Beglin, 1994). Linking back to restraint theory, the evidence now highlights that dieting does not necessarily lead to an individual becoming a restrained eater (Mills et al., 2021). This means that dieting and restrained eating should not be treated as the same construct. The construct of restrained eating, as measured by the restraint scale, does have supporting evidence that engaging in this type of restrained eating can lead to overconsumption and obesity. However, other constructs, which are not the same, do not support restraint theory. However, this could be because questionnaires such as the TFEQ and DEBQ are not measuring restrained eating and are instead measuring restrictive eating.

EBT Developed From Theory—Restraint, Disinhibition, and Susceptibility to Hunger (Three-Factor Eating Questionnaire)

Since the development of the restraint scale, other questionnaires have been developed that aim to measure different dimensions of eating behavior. The Three-Factor Eating Questionnaire (A. J. Stunkard & Messick, 1985) was heavily influenced by the concept of restrained eating and was constructed to measure three dimensions of eating behavior. As previously described, individuals with high levels of restrained eating (as measured by the restraint scale) consumed more of a test meal after consuming a preload than individuals with low levels of restrained eating (Herman & Mack, 1975; Herman & Polivy, 1980), and this behavior was termed "counter-regulation." Herman and Polivy (1983) proposed the boundary model of eating behavior, which can help to explain why high levels of restrained eating, as measured by the restraint scale, are associated with counterregulation after consumption of a preload. The boundary model suggests that eating is controlled by two boundaries along a continuum: the hunger zone and the satiety zone. The diet boundary (which lies between the hunger and satiety boundaries) represents a self-imposed, cognitive limit on food intake, designed to stop food intake before an individual reaches satiety. This model suggests that restrained eating requires cognitive motivation to limit food intake to reduce or control body weight. This definition of restrained eating also includes feelings of guilt after episodes of overeating. The initial restrained eating studies attempted to examine how restrained eaters behaved when presented with a disinhibitor, for example, alcohol (Polivy & Herman, 1976b), dysphoric emotions, for example, depression (Polivy & Herman, 1976a), and anxiety (Herman & Polivy, 1975). The results showed that these restrained eaters initially suppressed their intake through cognitive efforts but then surrendered to temptation and overeating, meaning restrained eating and disinhibition are both key concepts to restrained eating.

As mentioned in the Restrained Eating section, there are issues with the construct of restrained eating. However, A. J. Stunkard and Messick (1985) argued these limitations are with the scale used to measure restraint, not the concept itself, which led them to develop a new instrument to measure restrained eating and related issues, which was initially derived from the revised restraint scale (Herman & Polivy, 1980) and the Latent Obesity Questionnaire (Pudel et al., 1975). Additionally, 17 new items were created based on clinical experience. In their analysis, three factors were discovered and were interpreted as cognitive control of eating behavior, disinhibition of control, and susceptibility to hunger. Later analysis of the TFEQ by Westenhoefer et al. (1999) identified two dimensions of restrained eating, which were named flexible and rigid restraint. Rigid control was associated with higher disinhibition scores and BMI and more severe binge eating episodes, while flexible control was associated with lower disinhibition scores and BMI and less severe binge eating episodes. More recently, Westenhoefer et al. (2013) found that flexible restraint was associated with better weight loss maintenance, while rigid restraint was associated with less weight loss. However, other studies have failed to replicate these associations, which questions the utility of the distinction between flexible and rigid restraint. For example, Masheb and Grilo (2002) found that flexible and rigid restraint were significantly correlated with each other, and both negatively correlated with BMI in patients with binge eating disorder, which suggests that this distinction may not be useful for these patients.

There is evidence that the three EBTs measured in the TFEQ are associated with body weight and obesity. A literature review found consistent evidence that disinhibition is positively associated with BMI and fat mass (Bryant et al., 2019). However, the role of restraint using the TFEQ is also conflicting, whereby studies have found both positive and negative relationships with BMI. In a meta-analysis of the association between EBT, BMI, and energy intake, susceptibility to hunger and disinhibition were significantly and positively correlated with BMI and EI, with moderate effect sizes. The role of restrained eating using all available measures was again conflicting, as restrained eating was positively associated with BMI but negatively associated with energy intake (C. Dakin, Beaulieu, et al., 2023). Consequently, the mechanisms that underlie why there are differences in body weight and composition in individuals with different restrained eating and disinhibition scores are not clear but could be partially explained by the interaction between disinhibition and restrained eating and other factors such as eating patterns, diet quality, and eating responses to prompts and cues not measured by these scales (Bryant et al., 2019).

Restrained Eating, Emotional Eating, and External Eating (DEBQ)

The DEBQ, similarly to the TFEQ, was developed as another attempt to integrate constructs across different theories of eating behavior, including psychosomatic theory (see Psychosomatic Theory section), externality theory (see Externality Theory section), and restraint theory (see Restrained Eating section). Both psychosomatic and externality theories attribute the internal state (either emotional eating or a high degree of externality) as a causal factor in developing obesity. However, a major limitation of both theories is that a tendency to emotionally eat or a high degree of externality do not always lead to weight gain, and high levels of emotional and external eating can be found in all weight categories (Rodin, 1975, 1978). Restraint theory tried to account for these limitations (Herman & Mack, 1975; Herman & Polivy, 1980) and also suggested that external and emotional eating are consequences of dieting (Herman & Mack, 1975). All three theories have been tested; however, as previously described, there are limitations with each theory, including conflicting and inconsistent findings and a lack of replications.

The aim of the DEBQ was to develop a homogenous scale of restrained eating, emotional eating, and external eating to improve understanding of eating patterns in individuals with obesity. The authors identified that the TFEQ was a major improvement on previous scales. However, they noted that, at the time of their study, the TFEQ was not available. The DEBQ has been widely used since its development, and in general, the theoretical structure of this questionnaire has been confirmed (Cebolla et al., 2014). However, some items in the scales have been found to be problematic, including cross-loadings and some negligible loadings onto their proposed factors (Barrada et al., 2016; Cebolla et al., 2014). The implication of using the DEBQ with these problematic items is that they could be adding noise when using this scale for research or clinical analysis. In both studies, the authors suggest that deleting problematic items could increase the reliability of the scale and better distinguish between different dimensions such as boredom and emotional eating. Additionally, although the DEBQ has been widely used to assess self-reported emotional eating, multiple laboratory studies have found that emotional eating as measured by the DEBQ is unrelated to eating in response to emotional stimuli (Braden et al., 2020; Domoff et al., 2014). These findings suggest possible concerns with the validity of the DEBQ to measure the construct of emotional eating. Therefore, as previously suggested with the construct of restrained eating, the type of questionnaire used to measure emotional eating may be an important consideration for researchers.

Hedonic Hunger (Power of Food Scale)

Another development of an EBT from restraint theory is the concept of hedonic hunger (Lowe & Levine, 2005). In the literature, there is often a distinction made between physiological (the result of short-term energy reduction) and psychological hunger (driven by psychological rather than energy needs, e.g., eating when not hungry because of environmental cues to avoid negative emotions or simple pleasure). This dual-factor perspective is referred to as the standard model of hunger (Lowe & Levine, 2005). There is also a distinction in the brain between the homeostatic system, which is activated by energy deficits, and the hedonic system, which is activated by palatable food cues. Homeostatic appetite control is part of a psychobiological system that evolved to maintain a sufficient supply of nutrients for growth and maintenance. It includes both excitatory and inhibitory signals that influence appetite and food intake via tonic and episodic control mechanisms (Hopkins et al., 2017). In contrast, hedonic hunger is based on the availability and palatability of foods in the environment (Blundell & Finlayson, 2004). Food intake is dependent on the interaction between homeostatic appetite control and hedonic pleasure, which involve higher order processes including learning, memory, planning, and prediction. These processes generate not only the conscious experience of the sensory properties of food but also the hedonic pleasure that is acquired from the food (Kringelbach, 2004).

In the current obesogenic environment, there is a constant presence of palatable food cues, and this may chronically activate the hedonic appetite system. This further highlights that dietary restraint does not necessarily create counterregulatory responses solely induced by an energy deficit, and exercising restraint may be necessary to prevent weight gain and to promote weight maintenance (Schaumberg et al., 2016; Yeomans et al., 2004). Thus, as restraint theory suggests that restricting food intake below homeostatic needs produces deprivation (Herman & Mack, 1975), restricting palatable food intake could induce "perceived deprivation" even when an individual is in energy balance or in a positive energy balance (Lowe & Levine, 2005). The authors argue that motivation to eat more than needed is as powerful as the motivation to eat when deprived.

Evolutionary accounts suggest that it is adaptive to consume food in the absence of physiological hunger because it can prevent the onset of physiological hunger and increase storage of body fat which protects the body from periods of food scarcity (Peters et al., 2002; R. J. Stubbs et al., 2012). In resource-limiting environments, over the long term, hedonic needs safeguard the individual from unexpected changes in the environmental supply of energy nutrients. This concept indicates that food consumption can be driven by homeostatic needs (energy deprivation) or hedonic needs (presence of food, specifically palatable foods). The implication is that the rising rates of obesity are a consequence of passive overconsumption (Blundell & Gillett, 2001). However, eating for pleasure of other affective functions could also lead to active overconsumption. Thus, Lowe et al. (2009) developed the Power of Food Scale (PFS), a psychometric questionnaire which aimed to measure hedonic hunger. The PFS was designed to measure individual differences in appetite-related motivations, thoughts, and feelings in environments where palatable foods are available.

If the PFS measures a predisposition for overconsumption, a significant relationship between PFS and BMI would be expected. However, a clinical study found no statistically significant relationship between PFS and BMI (Cappelleri et al., 2009), and a meta-analysis of the PFS found no significant association between PFS and BMI (C. Dakin, Beaulieu, et al., 2023). In contrast, Vainik et al. (2015) did find a significant correlation between PFS and BMI and suggested that this association may be explained by an empirical two-factor solution of the PFS. The items that relate to uncontrolled eating have a positive association with BMI; however, the items that relate to food liking either have no association or a negative association with BMI. Potentially, the lack of association between PFS and BMI could be because studies assess total PFS responses, meaning they do not split PFS items into their two-factor solution.

Intuitive Eating (Intuitive Eating Scale)

A last EBT to be discussed that has been developed from psychological theory is intuitive eating. From a psychological perspective, the study of eating behavior has focused on disordered rather than healthful types of eating. However, there is also a need to identify and study healthful eating behaviors that help to maintain psychological health (Seligman & Csikszentmihalyi, 2000). One example of a healthful EBT could be restrained eating. However, as previously described, there is mixed evidence about whether high levels of restrained eating lead to positive or negative health outcomes. Restraint theory argues that counterregulatory responses can lead to disinhibited and binge eating behavior (Herman & Mack, 1975; Herman & Polivy, 1980), while other researchers suggest that demonstrating flexible control when eating is associated with weight loss and weight loss maintenance (M. T. McGuire et al., 1999; Phelan et al., 2009). Furthermore, healthful eating has been discussed within other disciplines but is often considered regarding guidelines for intakes of specific foods (Ogden, 2011). As a result, there is less known about healthful eating behaviors (Tylka, 2006).

Healthful eating is important to study because it is more than just the absence of disordered eating behaviors; internal cues can be used to determine behavior. For example, healthful eating involves using physiological hunger and satiety cues as a guide to determine when, what, and how much to eat (Tribole & Resch, 1995). Consequently, healthful eating is likely to be negatively correlated with an absence of eating disorder symptoms, but it is not only defined by this absence. This led Tylka (2006) to develop a questionnaire to assess the psychometric properties of potential variables that can serve to protect against developing disordered eating behaviors. The focus of this questionnaire was to describe and measure intuitive eating, which is defined as eating based on physiological hunger and satiety cues, rather than external and emotional cues (Tribole & Resch, 1995). Intuitive eating can be viewed as a healthful eating behavior because it is strongly related to understanding and responding to internal physiological needs of hunger and satiety, in addition to a lack of concern with food. In the literature, three central dimensions of intuitive eating have been identified, and in the initial intuitive eating questionnaire (Intuitive Eating Scale [IES]), these three dimensions were found and given the following labels: unconditional permission to eat, eating for physical rather than emotional reasons, and reliance on hunger and satiety cues.

Research has found support for the construct validity of the original IES, including finding negative correlations with total intuitive eating scores and disordered eating symptoms (Shouse & Nilsson, 2011) and BMI (Augustus-Horvath & Tylka, 2011). However, the original scale omits one of the components of intuitive eating proposed by Tribole and Resch (2012), which involves honoring health or engaging in gentle nutrition (choosing foods that promote health, energy, stamina, and body performance). If intuitive eating is a healthful eating behavior, this construct should also reflect this dimension of choosing nutritious foods that help the body to perform well. Therefore, the Body-Food Choice Congruence subscale (B-FCC) was added to the Intuitive Eating Scale–2 (IES-2; Tylka & Kroon Van Diest, 2013). The new IES-2 was found to improve upon the original IES, and the B-FCC was found to be a distinct factor within the IES-2. More specifically, B-FCC was significantly correlated with specific body-related measures and all psychological well-being measures. Additionally, B-FCC predicted unique variance in self-esteem, positive affect, negative affect, and life satisfaction above and beyond the variance accounted for by eating disorder symptomology. Last, the addition of B-FCC supports the interpretation that components of intuitive eating are associated with listening to and appreciating the body.

Furthermore, another similar model to intuitive eating is the scatter eating competence model (eSatter; Satter, 2007). Although this model examines a wider spectrum of eating attitudes and behaviors than the intuitive eating scale, eSatter highlights the importance of intuitive eating. According to the model, a key construct of a competent eater is "having internal regulation skills that allow intuitively consuming enough food to give energy and stamina and to support stable body weight." The author also proposes, similarly to Tylka (2006), that it is important to study this healthful eating behavior because it will not only allow for more targeting interventions but also help individuals to trust their own capabilities to learn and develop a more adaptive eating style.

Dual Process Model (Social Behavior Theory)

Separate from psychological explanations of eating behavior, Strack and Deutsch (2004) argued that social behavior is controlled by two systems: reflective and impulsive processes that can operate in accord or compete with each other. While not specifically aimed to explain eating behavior, this theory can be applied to eating. For example, Finlayson et al. (2007) discussed the implications of a dual-process perspective of food reward for weight gain and obesity. Behavior in the reflective system occurs as a consequence of a decision process, requiring high cognitive capacity to enable cognitive reasoning. Distraction and low levels of arousal will interfere with the reflective system. In contrast, the impulsive system operates through spreading activation; it is fast, inflexible, and needs no attentional resources. Consequently, processes in the reflective system are interrupted more easily than processes in the impulsive system.

Both systems lead to the activation of behavioral schemata but differ in how they activate a behavioral schema (Strack & Deutsch, 2004). Conflicts can arise if the activated behavior schemata are incompatible. These conflicts can be seen in eating behavior. For example, the sight of food, which acts as a food cue, could activate the impulsive system, which may trigger a desire to eat or habitual food intake (Schüz et al., 2015). Indeed, it is well documented that eating behavior is often driven by the response to food-related cues (Lowe & Butryn, 2007). However, the reflective system could also be activated, and depending on the individual's current state (e.g., level of hunger/satiety, motivation to eat; Cheon et al., 2019) or traits (e.g., whether the individual engages in restrained eating or mindful eating; Polivy et al., 2020; Warren et al., 2017), or motivations, the individual may consciously decide they do not want to engage in an eating episode.

This means that the impulsive system is in conflict with the reflective system about whether or not to engage in eating. According to the dual-process model, the resolution of the conflict between the reflective and impulsive systems depends on the strength of the activation for each schema. For example, if the food is not well liked by the individual, then the eating schema may not be highly activated, as the individual may not strongly desire the food. Whereas, if an individual is distracted or under high levels of stress, they may not have the cognitive capacity available to highly activate the reflective system, and therefore the impulsive system may prevail.

Strack and Deutsch (2004) also noted that while both systems contribute to behavior, the impulsive system will assume primary control if operating conditions for the reflective system (e.g., cognitive capacity) are not fulfilled. This means that eating behavior is less likely to be determined by conscious control over food intake (assessing the valence and probability of future consequences of the behavior) and more likely to be determined by immediate associations that result in hedonic qualities. Indeed, stress has been shown to interfere with cognition (Tomiyama, 2019) and lead to unhealthy eating (Raio et al., 2013). Some individuals who report high levels of perceived stress show increased disinhibition, binge eating, and emotional eating compared to individuals reporting lower levels of perceived stress (Diggins et al., 2015; Groesz et al., 2012). These eating behaviors could reflect the impulsive system taking control, as disinhibition, binge eating, and emotional eating focus on central features of the impulsive system, including emotions, desires, and habits controlling eating behavior.

The connection between stress and overeating may not always be automatic, as some people may use food as a deliberate coping strategy for stress. Researchers do not often ask participants directly about their use of health behaviors, for example, eating, as coping responses to stress (Park & Iacocca, 2014). However, a poll of 1,420 American adults asked participants how they cope with stress, and among the top answers was eating (American Psychological Association, 2012). Additionally, a focus group of low-socioeconomic status individuals found that some individuals reported eating a poor diet in response to stress to "self-medicate" (S. A. Kaplan et al., 2013). This indicates that, in some cases, eating in response to stress is deliberate, meaning stress may influence both the reflective and impulsive systems. However, in this focus group, participants also reported that their willpower to resist health behaviors (e.g., eating) was reduced after a long and stressful day. These findings indicate that while eating in response to stress can be a deliberate coping strategy, it is also an automatic response, especially when control/reflective resources are low (e.g., after a long and stressful day). Indeed, although many individuals report eating more than usual to cope with stress (S. A. Kaplan et al., 2013), research also demonstrates that many individuals are not aware of their motives for engaging in health behaviors (e.g., eating; Sheeran et al., 2013). DeSteno et al. (2013) suggested that eating serves an "implicit" emotion-regulation strategy, which means that engaging in these behaviors does not meet the commonly used definition of coping as deliberate.

Overall, dual-process theory was developed to explain social behavior; however, many researchers have applied this approach to eating behavior, and it is now well recognized that motivations for eating are influenced by two major systems: conscious and automatic processes. Behavioral research supports the concept of a dual-process model of obesity, which suggests that a discrepancy between conscious and automatic processes influences eating behavior, which promotes a positive energy balance and obesity (Strack & Deutsch, 2004). The field of behavior change also supports the concept of conscious and automatic components that influence weight management behaviors (Greaves et al., 2011). Furthermore, previous analyses of EBTs on weight change during an 18-month weight maintenance trial found that increases in reactive eating (automatic processes) and decreases in reflective eating (conscious processes) were significantly and independently associated with concomitant weight change (C. A. Dakin, Finlayson, et al., 2023).

Other Psychological Theories Related to Dual-Process Theory

Since the development of the dual-process model of social behavior, other theories of eating behavior have emerged that attempt to explain motivations for eating, overeating, and obesity, as well as other types of eating patterns, for example, restrained eating and binge eating. Similar to dual-process models, many of these theories regard eating patterns as the results of two competing processes. For example, in the dual pathway model of overeating, restrained eating and negative affect compete to control eating behavior and are thought to lead to bulimia nervosa (Ouwens et al., 2009). The restrained eating pathway can be seen as a form of reflective process, as it is a conscious attempt to restrict eating that involves deliberate cognitive reasoning. Conversely, the second pathway, negative affect, suggests that people with bulimia use purging and bingeing as a means of regulating negative emotions and mood states. Individuals who eat in response to distress could be confusing emotional distress with hunger due to a lack of interoceptive awareness, or they may learn to use eating as a means of comforting negative affect (Van Strien et al., 2005). The measures used to assess negative affect include emotional eating and drive for thinness, which involve more automatic processes and, therefore, can be seen as similar to the impulsive system.

Another later developed model of eating behavior, the goal conflict model (Stroebe et al., 2013), suggests that the difficulties restrained eaters have in regulating their food intake occur due to a conflict between two incompatible goals: eating enjoyment and weight control. This model proposes that individuals fail to regulate their intake due to palatable food cues in food-rich environments that strongly prime the goal of eating enjoyment. Subtle reminders of food/eating trigger processes that activate this goal outside of conscious awareness and lead to inhibition of the weight control goal, which hinders healthy eating (Kavanagh et al., 2005). On the other hand, extended priming of the weight control goal, which is defined as reaching and maintaining a target weight, leads to inhibition of the eating enjoyment goal through deliberate eating behavior such as dieting.

In the current food-rich environment, it is likely that the eating enjoyment goal is more strongly activated than the weight control goal because individuals are constantly surrounded by palatable food cues. This can often lead to dieting failure and weight relapse in many people engaged in restrained eating for the purposes of weight management. Evidently, there are similarities between the goal conflict model and dual-process models. Both theories suggest that resources are important, proposing that individuals are successful in maintaining behavior if they have enough psychological and physical resources. Dual-process models hypothesize that behavior can be controlled by the resource-intensive reflective system or the automated impulsive system. Resources can act as moderators and determine which system generates a response (Kwasnicka et al., 2016). Studies have found that when control resources were available to regulate the goal of dieting/weight control, explicit measures, such as the participant's intention to diet, predicted consumption. However, when resources were low, implicit measures related to the hedonic relevance of food stimuli were better predictors of consumption (Friese et al., 2008). However, and importantly, many of the empirical findings related to dual-process models were tested and published before the replication crisis (Bem, 2011). Recently, Sommet et al. (2023) conducted a metastudy of 159 studies from influential psychology journals and found that the median power to detect interactions of a typical sample size was 0.18. Given that typical psychological studies, including those described above, have only 0.18 power to detect these findings, caution must be taken when interpreting results and when considering the validity of dual-process findings.

Behavioral Susceptibility Theory

Behavioral susceptibility theory (BST) combines evidence from both genetics and the role of the environment to better understand how appetite influences obesity onset (Llewellyn & Wardle, 2015). This section describes the theory as it relates to the psychology of eating behavior. BST argues that individual differences in appetite play a causal role in the development of obesity, suggesting that those who inherit lower sensitivity to satiety or an increased avid appetite are more likely to overeat in response to the food environment. The theory suggests that the two important components of appetite are responsiveness to hunger cues and responsiveness to satiety cues. Those who are more responsive to food cues are more likely to eat in response to an attractive food opportunity. Additionally, those who have weaker internal satiety cues or are less aware of their satiety cues are more likely to eat for longer. Where this theory differs from previous similar theories, for example, externality theory, is the suggestion that genetic risk operates through these two appetitive traits, and this results in differences in susceptibility to the food environment. This means that obesity genes determine each person's responsiveness to food opportunities, and as opportunities to eat increase, genes for high food responsiveness and low satiety responsiveness are expressed in terms of behavior. Consequently, BST aims to provide a better explanation for how genetics, the environment, and appetite interact.

The two new constructs that BST presents are food responsiveness (FR) and satiety responsiveness (SR), which can be measured in adults using the adult eating behavior questionnaire (AEBQ; Hunot et al., 2016) or the child eating behavior questionnaire (Wardle et al., 2001). In support of BST, research has found that increased FR and decreased SR are EBTs that are associated with greater adiposity (Carnell & Wardle, 2008). Additionally, both EBTs are linearly associated with weight across the entire weight spectrum (from underweight to obesity in children; Croker et al., 2011). Furthermore, a longitudinal study has found that these EBTs are more likely to drive early weight gain than early weight gain driving changes in EBTs (van Jaarsveld et al., 2011). Overall, BST has built on previous theories that indicate the important role of appetite in obesity, which address how appetite, genetics, and the environment all play a causal role in obesity development.

Frameworks That Attempt to Integrate Constructs From Psychological Theory

The suggestion that many EBTs share similar underlying constructs has been previously proposed by Vainik et al. (2015). It is logical to recognize that if EBTs share underlying constructs, then there should be an overarching theoretical basis to which those underlying constructs may belong. The authors suggest that EBTs could be capturing the same underlying construct, termed "uncontrolled eating," which is defined as increased appetite and decreased self-control. EBTs could also be measuring different levels of severity along this construct. Vainik et al. (2015) adapted a continuum model of uncontrolled eating proposed by Davis (2013) to incorporate five EBTs into this model. They suggested that EBTs, including eating impulsivity, hedonic hunger, emotional eating, disinhibition, and binge eating, measure different ranges of values along a broader continuum of uncontrolled eating.

The continuum begins at "homeostatic eating," where an individual's energy intake matches their energy expenditure. Early stages of uncontrolled eating are triggered by responsiveness to food cues in the environment, resulting in "passive overeating." This reflects a limited capacity to recognize a positive energy balance, and therefore an individual exerts little control over their eating behavior (Prentice & Jebb, 2003). More severe stages of uncontrolled eating then follow, which include emotional eating, disinhibition, and binge eating. Using bifactor analysis and item response theory, the authors found evidence to support the uncontrolled eating model, which suggested that these five EBTs capture the same common construct of uncontrolled eating and each focus on different severities of uncontrolled eating. This novel research provides a starting point to develop a comprehensive perspective of the current literature, which suggests that EBTs are measuring different degrees of lack of control of eating. However, only women were included in the analysis, and a limited number of EBTs were assessed, which were all constructs that measure moderate to severe stages along the continuum of uncontrolled eating.

Racine et al. (2019) examined a wider variety of nonhomeostatic EBTs, which included seven questionnaires (DEBQ, Eating Pathology Symptoms Inventory, Loss of Control Over Eating Scale, Binge Eating Scale, EDE-Q, PFS, and Yale Food Addiction Scale). Their results supported a seven-factor solution, suggesting that the items in these questionnaires measure seven multiple but related constructs termed emotional eating, loss of control over eating, overeating, distress over nonhomeostatic eating, hedonic hunger, and food addiction. These results are in contrast to Vainik et al. (2015) and suggest that the questionnaires used are measuring multiple constructs. Although there is still a lack of research that examines the less severe end of the uncontrolled eating spectrum, or "homeostatic eating" constructs. Consequently, there is a need to examine whether a framework can be developed that explains more varied EBTs, including EBTs from both ends of the homeostatic eating/uncontrolled eating spectrum.

Furthermore, components of behavior change models and cognitive theories can also be used to explain how individual differences in eating behaviors can help or hinder weight management attempts. These models also draw similarities with dual-process models. For example, models that explain behavior change during weight management attempts also include reflective and impulsive components that work against each other to influence changes in eating behavior that influence weight change (Dunton et al., 2021; Greaves et al., 2011; Kwasnicka et al., 2016). Greaves et al. (2011) suggested that longer term weight management involves tension between existing habits (EBTs) and incompatibility of new weight management behaviors. Research has found that engaging in reflective and impulsive processes can influence weight management success. Engaging in and/or developing reflective processes such as self-regulation, motivation, and managing external influences (Dombrowski et al., 2014; Teixeira et al., 2005; Varkevisser et al., 2019) has been found to aid weight management. In contrast, reactive/automatic processes (e.g., emotions and desires) are the result of associative learning and physiological resistance to weight loss, which can undermine longer term weight loss (Berthoud, 2011; Blundell & Finlayson, 2004).

Additionally, the capability, opportunity, motivation–behavior (COM-B) model (Michie et al., 2011) was developed as a framework of behavior change that involves three essential conditions: capability, opportunity, and motivation. The framework took a systematic approach to behavior change with the aim of capturing a range of mechanisms, including the internal and external environment. Capability is defined as an individual's physical and psychological capacity to undertake a specific activity. Motivation is defined as any brain process that directs behavior. This involves goals, conscious decision making, habitual processes, and emotional responding. Last, opportunity is defined as any factor outside of an individual that makes a behavior possible. The COM-B model differentiates between reflective processes and automatic processes, which further highlights similarities with dual-process models. The COM-B model also draws concepts from dual-process models (Strack & Deutsch, 2004) to help with linking interventions to different components in the behavior system. For example, the COM-B model also suggests that automatic processing is central to the behavior system and should be considered equally to reflective, systematic, and cognitive processes when designing interventions. Overall, COM-B makes reference to many EBTs and dimensions covered by psychological theory and attempts to integrate them to predict behavior. However, the COM-B model refers to social behavior and not eating behaviors specifically. Therefore, it would be useful to develop a comprehensive framework of EBTs to understand the dimensions/constructs that underlie eating behaviors.

Limitations of Current EBTs and Frameworks

The current state of EBTs presents some notable critiques. First, there is an abundance of EBTs, which has led to a lack of clarity in understanding the underlying mechanisms of an EBT. The extensive developing of new traits has resulted in a convoluted array of measures without meaningful organization. This means there is a lack of understanding about the overarching domains of eating behavior that EBTs are measuring. Second, many EBT constructs appear to be closely related or redundant, which has also led to conceptual overlap and confusion. For example, there are more than four different scales used to measure restrained eating, for example, the EDE-Q, restraint scale, TFEQ, and DEBQ (Fairburn & Beglin, 1994; Herman & Mack, 1975; A. J. Stunkard & Messick, 1985; Van Strien et al., 1986). Each scale is proposed to measure restraint; however, they are not measuring the exact same construct, which is termed a jingle fallacy (Thorndike, 1904). The restraint scale appears to measure excessive overconsumption, weight fluctuations, and cognitions and emotions in addition to dietary restraint (Polivy et al., 2023). In contrast, the TFEQ primarily identifies successful dieters who can restrict their intake. A jingle fallacy has also been found in the personality literature, where although some evidence suggests overlap between individual traits and values, the current body of evidence indicates that personality traits and values are distinct constructs (M. Higgs & Lichtenstein, 2010). Furthermore, different scales used to measure restrained eating have found conflicting evidence about the role of restraint in overeating and obesity. This indicates how vital it is for researchers to understand what exactly the scale they are using measures because important outcomes of a study could be influenced by the EBT measure used. Similarities between constructs not only add to the complexity of the field but also limit the ability to distinguish unique contributions from each EBT.

A comprehensive framework that encompasses the various dimensions of eating behavior and provides a coherent structure for the research and application of EBT measurements is currently lacking. To advance the field and provide more meaningful insights into eating behavior, researchers should aim to consolidate the description of EBTs, identify meaningful relationships among EBTs, align EBT measures with existing theory, and work toward developing a comprehensive framework that identifies the underlying common constructs that are shared between overlapping EBT measurements.

Summary of Psychological and Cognitive Theories

In this section, key psychological and cognitive theories, EBTs developed from theory, and attempts to develop frameworks that integrate constructs were reviewed. Their contribution to understanding eating behavior and obesity were evaluated (see Table 1 for a summary). Overall, many of these theories draw on concepts from dual-process models, which suggest that eating behavior is driven by two competing processes. Additionally, many theories draw on approach versus avoidance-based traits, internal versus external traits, and implicit versus explicit traits. The main dimensions that seem to be captured by many theories and often EBT questionnaires seem to focus on one or two of the two competing processes that drive eating behavior. One process involves cognitive, reflective, and decision-making abilities, which include sensitivity and responsivity to internal cues. The other process involves automatic and reactive behavior, which includes reacting to external cues in the environment, eating for pleasure/comfort, and overeating to reduce negative or enhance positive emotions.

Although these theories and models are presented as separate, it is important to highlight that there is considerable overlap between each theory/model and many of the newer theories incorporate concepts from earlier theories to develop a more comprehensive understanding of eating behavior. Consequently, it is suggested that there is an overarching theoretical umbrella to explain human eating behavior, whereby many EBTs fall under two competing processes. The next section now reviews physiology and neurobiology with the aim of integrating these psychological and psychometric aspects of behavior into the current understanding of the biology of eating behavior.

Physiology of Eating Behavior

Human appetite, or motivation to eat, is an example of a phenomenon that is biopsychological in nature. As discussed above, there are psychological and cognitive influences on eating behavior, and many models describe elements of dual-process theory, whereby decisions are motivated by separate cognitive (slow and reflective) and affective (fast and reactive) networks. However, many of these theories and models oversimplify the interplay between cognitive, affective, and metabolic systems as well as the neuroanatomical circuitry that links these systems into patterns of goal-oriented, motivated behaviors (Keren & Schul, 2009). Consequently, integrating knowledge of physiology and neurobiological processes that signal motivation to eat (food seeking), appetite behaviors (consummation), and satiation/satiety (cessation of eating) within the current energy balance framework could offer a more detailed and comprehensive understanding of the role of EBTs in appetite research.

There are short- (episodic) and long-term (tonic) physiological circuits influencing of motivation to eat and eating behavior (Blundell et al., 2001). Episodic processes occur periodically, are organized around bouts of eating (e.g., meals or snacks), and typically refer to internal signals from the gut, mediating satiation and satiety. These signals act via the gut–brain axis and feed into the same hypothalamic circuitry as tonic signals. In contrast, tonic processes change over longer periods of time more relevant to energy balance regulation and signal changes in body size and composition. This long-term appetite control involves the secretion, inter alia, of insulin by the pancreas and the secretion of leptin by adipocytes in proportion to fat mass (Wynne et al., 2005). Both insulin and leptin act on receptors in the hypothalamus, which excite inhibitory neurons that decrease food intake and inhibit excitatory neurons that increase food intake. The overall net balance of these signals is believed to control the drive to eat.

Traditional models of homeostatic appetite control argue that adipose tissue is central to appetite control and is the main driver of food intake, with leptin also playing a key role in appetite control (Woods & Ramsay, 2011). However, Blundell et al. (2015) demonstrated that fat mass is weakly and negatively associated with food intake in people with obesity, whereas levels of fat-free mass (FFM) and RMR are positively correlated with hunger and energy intake and could be tonic signals of energy need and motivation to eat. Indeed, Blundell et al. (2020) considered FFM and RMR to be major determinants of energy intake in humans, and since 2011, the relationship between FFM, RMR, and EI has been substantially replicated in more than 12 studies from more than seven different countries. Integrated signals from changes in body size and composition influence the leptin/insulin and related neuropeptide axes to defend against significant weight loss. However, in most individuals, these signals are less responsive to weight gain under modern environmental circumstances.

One of the key theories in the field of human appetite suggests that energy balance is biologically regulated. The concept of energy balance regulation is referred to as homeostasis. Cannon (1929) first defined homeostasis in the context of appetite and suggested that under normal circumstances, variations from the mean of regulated variables would impair the functions of a cell or organism, and therefore, before any extremes are reached, responses are automatically produced that act to bring the disturbed state back to the mean. This definition implies that the body's internal environment is maintained within a controlled homeostatic range. Since the late 20th century, numerous theories have been developed that propose that almost all components of the energy balance system can function as a negative feedback homeostasis signal to influence energy intake. These theories include the previously mentioned models that propose that appetite is regulated by adipose tissue (Woods & Ramsay, 2011) or FFM and RMR (Blundell et al., 2015). In addition, temperature regulation (Brobeck, 1946), fat (Kennedy, 1953), carbohydrates (Mayer, 1953), amino acids (Mellinkoff et al., 1956), protein leverage (Raubenheimer & Simpson, 2019), and body weight (Hervey, 1969) have all been suggested as homeostatic signals that influence energy intake.

Observational evidence for homeostatic regulation of body weight comes from the finding that a large proportion of the population is able to keep a stable body weight over time, which implies the existence of active regulation or defense (Shin et al., 2009). It has also been argued that such apparent stability could simply be a function of the complexity of multiple redundant systems. This is evidenced in human and rodent under- and overfeeding studies where weight loss or gain is rapidly corrected by compensatory increases and decreases in calorie intake (Hall, 2006; Keesey & Corbett, 1990; Siervo et al., 2008). However, many of these studies test participants over durations of less than a few weeks and involve extreme overfeeding regimes. Furthermore, the prevalence of obesity has called into question whether body weight is regulated because "if body fatness is under physiological control, then how come we have an obesity epidemic?" (Speakman, 2014, p. 88). Additionally, Watts et al. (2022) argued that the concept of set point for body weight is not consistent with the majority of experimental findings. To account for this, Wirtshafter and Davis (1977) argued for a simple feedback control model that does not include a set point but can account for evidence that does support body weight set point existence. They coined the term "settling point," which suggests that body weight is regulated, but this regulation can vary considerably. An alternative proposal is that regulation of body weight is asymmetric (Blundell & Hill, 1985). The balance of evidence suggests that energy balance is regulated, but the regulation is not precise or symmetric. While there are physiological changes that oppose weight gain or loss, excess energy intake is tolerated (weak episodic inhibitory control and no tonic negative feedback for weight gain), whereas energy deficit is strongly opposed (strong periodic and tonic negative feedback signals for energy need; R. J. Stubbs & Tolkamp, 2006; R. J. Stubbs & Turicchi, 2021). This means that satiety can easily be overridden, which allows for positive energy balance (Blundell & Macdiarmid, 1997).

In the energy balance model, overconsumption is viewed as being predominantly influenced by highly palatable, energy-dense foods (Hall et al., 2022). The asymmetry of body weight regulation is likely to have evolved because a positive energy balance is much more adaptive for survival than a negative energy balance (Speakman, 2018). Additionally, evidence suggests that the level of feedback is not linear. This means that small deviations from current body weight will induce virtually no or weak compensatory responses. As deviations increase, there will be far greater compensatory responses. In particular, compensatory responses are more severe in response to large negative rather than positive energy balances over time (R. J. Stubbs et al., 2004). Taken together, energy balance is regulated, but this regulation is asymmetric and imprecise. Furthermore, physiological and psychological responses to weight loss occur on a continuum, and this is influenced by the amount of energy deficit, the duration of energy deficit, body composition at the beginning of the deficit, and the psychosocial environment in which the energy deficit occurs (R. J. Stubbs & Turicchi, 2021).

Taking a physiological perspective on appetite control and energy balance, it becomes apparent that there are homeostatic processes manifesting in physiological hunger that also mediate the strength and duration of satiety responses. These processes can help to better explain eating behavior and therefore constitute traits that should be captured in a framework of eating behavior. There already exists some EBTs that could be used to measure physiological hunger. For example, the hunger subscale of the AEBQ (Hunot et al., 2016) is a measure of physical hunger that includes items such as "I often notice my stomach rumbling." Evidence shows that physiological signals, including levels of FFM and RMR, are positively correlated with hunger and energy intake and act to motivate eating (Blundell et al., 2015). Therefore, it is important for EBTs to distinguish between physiological hunger and hedonic hunger as measured by the PFS. Indeed, another subscale in the AEBQ measures responsiveness to food, which can be seen as a proxy of hedonic hunger, and although these scales were strongly related to each other, a confirmatory factor analysis revealed a better model fit when these scales were kept separated. Furthermore, there is good evidence in the literature that these scales measure distinct dimensions of eating (Schachter, 1968; A. J. Stunkard & Fox, 1971).

There are also EBTs that could be considered to measure homeostatic processes. For example, the satiety responsiveness scale from the AEBQ (Hunot et al., 2016) measures an individual's sensitivity to their internal levels of satiety. The concept of satiety responsiveness is also represented in the internal hypothesis of externality theory (Schachter, 1967). The internality hypothesis suggests that eating behavior is influenced by internal physiological cues that are associated with food deprivation. Taking an energy balance perspective, internality or satiety responsiveness measures how well an individual responds to internal feedback so that energy intake and energy expenditure are in balance. Clearly, physiological hunger and homeostatic processes are related; hunger is influenced by differences in interoceptive awareness, the strength of satiety signals, and the drive of energy requirements. Consequently, it is important to develop a framework of EBTs that can measure both sides of the energy balance equation as well as other psychological traits that influence eating behavior. EBTs that are currently available are limited in that they measure more of certain types of eating behavior and less of others. There are a limited number of EBTs that could be measuring physiological hunger or homeostatic eating. It would be valuable to develop new scales that cover gaps, for example, more specific measures of physiological aspects of eating behavior.

Neurobiology of Eating Behavior

While appetite is regulated by processes of hunger and satiety, there are many other drivers of eating, including cognitive and psychological drivers (as previously described) and food reward (Berthoud & Morrison, 2008). Neurobiological studies have been able to give a more comprehensive account of how hedonic, reward-based pathways are more important to energy balance homeostasis than has been previously suggested. Eating is a highly salient source of reward and pleasure for animals and humans. Pleasure is a central cue that links food liking and wanting to learned consumption behavior (Kringelbach et al., 2012). This suggests that human eating behavior is not governed solely by homeostatic mechanisms, and instead, pleasure and reward also play a central role in control or lack of control of energy intake (Kringelbach, 2004). Indeed, consuming palatable foods engages systems that are involved in reward processes, including dopamine and opioid systems (Pfefferbaum et al., 1992). Neurobiological studies can also help to explain why eating behavior so often leads to a positive energy balance in the current environment where several facilitatory cues and a lack of constraints are present concerning what, when, and how much to eat.

It is possible that in resource-limiting environments from which we evolved, homeostatic and hedonic systems operated in a coordinated manner to facilitate overconsumption during periods of food abundance. Natural selection would favor these behaviors due to environmental uncertainty (Berridge & Kringelbach, 2008; R. J. Stubbs & Tolkamp, 2006; R. J. Stubbs & Turicchi, 2021). However, modern environments have changed very rapidly compared to the environments that influenced energy balance regulation. The rate of change of the modern food environment is far faster than the rate at which human evolution can adapt. This means that at times, motivation to eat and food reward behaviors conflict with other non-food-related motivations and rewards, for example, health and well-being. Consequently, understanding how both homeostatic and hedonic factors influence food intake in the current environment requires knowledge of physiology, psychology, and neurobiology, as well as an appreciation of the asymmetry of energy balance regulation.

Motivation and emotion both drive food intake, which is supported by reward and hedonic processing (Kringelbach et al., 2012). Motivation to eat can be triggered by many factors, including metabolic need and hedonic drives, and there are numerous neurocircuits that underlie these processes. Physiological signals can modulate processing of cognitive and reward functions, which can affect regulatory processes. On the other hand, cognition and emotions can also influence homeostatic regulation, which can lead to energy imbalance. Appetite is therefore determined by the interaction between homeostatic and hedonic pathways (Berthoud, 2011). Using a neurobiological model of eating behavior that operates in the context of asymmetric energy balance regulation, many factors can largely influence motivation to eat, eating behavior, and energy balance (Berridge & Kringelbach, 2008; Kringelbach et al., 2012; R. J. Stubbs & Tolkamp, 2006). These models can help to explain the development of obesity in the current food environment as well as the difficulties people face when trying to lose weight and maintain weight loss.

As neurobiology can give a more comprehensive understanding of eating behavior, it is important that neurobiological mechanisms are captured by EBTs. Examples of neurobiological mechanisms that could be captured by EBTs include food reward, executive function, affect-related eating, and stress-related eating. Historically, there has been a focus on how emotions can affect eating behavior. Positive emotions (e.g., happiness) and negative emotions (e.g., stress and anxiety) can motivate eating through positive reinforcement and negative reinforcement of the emotional state (Skinner, 1963). This is also highlighted in psychosomatic theory, whereby overeating is thought to be motivated by drive reduction. Eating is associated with a reduction of negative emotions, and thus eating is rewarded (H. I. Kaplan & Kaplan, 1957). Emotional eating has been widely studied, and many EBTs have been developed that measure some aspect of emotional eating, for example, emotional eating (DEBQ), positive emotional eating (Positive and Negative Emotional Eating Scale), negative emotional eating (Positive and Negative Emotional Eating Scale), emotional overeating (AEBQ), and emotional undereating (AEBQ).

Furthermore, as previously described, food reward plays an important role in eating behavior, with some individuals displaying strong drives to eat in response to the natural reward of highly palatable foods (Beaver et al., 2006). This concept has been referred to as reward-based eating, and there is evidence for reward-based eating in both behavioral and neurobiological studies (Volkow et al., 2012, 2013). Researchers have proposed that reward-based eating can lead to overconsumption because it overrides satiety signals. The excess energy intake from overconsumption is reinforced over time through changes in dopaminergic pathways that regulate the neuronal systems related to reward sensitivity (Volkow et al., 2011). Measuring reward-based eating using an EBT would be useful to further understand how food reward influences eating behavior because this pattern of eating could lead to obesity (Epel et al., 2014). The PFS does tap into aspects of reward-based eating but focuses more on the impact of the food environment. Consequently, the reward-based eating drive (RED) scale was developed as an EBT to measure reward-based eating (Epel et al., 2014). In support of reward-based eating, the RED scale was found to be positively corelated with BMI. Since its development, the RED scale has been broadened to include additional items to better assess the entire spectrum of reward-related eating. Most recently, the Reward-Based Eating Drive–13 scale has been developed, which was also found to be positively correlated with BMI as well as Type 2 diabetes and craving for sweet and savory foods (Mason et al., 2017).

There is also evidence that higher level cognitive functions, including learning and memory, influence appetite control and that homeostatic signals also influence cognition (S. Higgs et al., 2017). Decisions around daily eating vary in terms of the cognitive effort that is required to complete them. In general, the automaticity of reward-driven decision making encourages habits that lead to consumption of energy-dense, highly palatable foods (Iso-Ahola, 2017). Contrastingly, decisions that sustain healthy eating behaviors require more effortful and conscious control, which is termed "executive function" (J. T. McGuire & Botvinick, 2010). Executive function is a limited-capacity resource; therefore, depending on the cognitive resources available and the current situation, cognitive or affective neural systems are activated differently to influence eating behavior decisions. Additionally, research shows that a high-fat, high-sugar diet can damage the hippocampus and decrease learning and memory in rodents (Davidson & Stevenson, 2022). Furthermore, reduced executive function renders individuals more susceptible to reactive processes, which influences food reward-based decision making and could lead to food seeking and, consequently, obesity (Zhou et al., 2023).

It is therefore apparent that both cognition and executive function also influence eating behavior. At present, there are no EBTs that are designed to measure cognitive-related eating or executive function related to eating. There are several food-related executive functioning tasks. For example, a systematic review identified 66 different neurocognitive tasks, of which 21 tasks measured executive function in relation to obesity and eating behaviors (Vainik et al., 2013). The review found that tasks that were sensitive to executive function and food motivation provided the most robust and reliable associations with BMI, BMI change, or eating behaviors. However, out of the 66 tasks reviewed, fewer than 11% showed consistent and reliable effects. Additionally, while these tasks are related to food and eating, for example, the food stroop task, food delay discounting, and relative reinforcing value of food, they are less relevant to self-report measures of eating behavior. As such, many of these tasks would not be considered EBTs, which are self-report measures of eating behavior. This discussion highlights areas for methodological development, including the development of EBTs that specifically measure executive functioning related to eating behavior.

Interestingly, within a neurocognitive model of appetite control, the processes of reflective and impulsive eating can be seen. Indeed, Herman and Polivy (2014) suggested that humans do not consciously consider food decisions all of the time, and instead, most of the time, eating engages no mental effort and is "mindless." The automatic food decisions that are made mirror the reactive processes within appetite control, which are suggested to be in control, especially when cognitive resources are low. Zhou et al. (2023) also argued that motivation to eat and food reward systems are mainly subcortical and reactive, while decision making involves conscious cortical processing and complex motivation. Thus highlighting the two competing pathways: impulsive and reflective eating. See Figure 1 for a graphical representation of an integrated physiological, neurobiological, and psychological model of energy balance regulation.

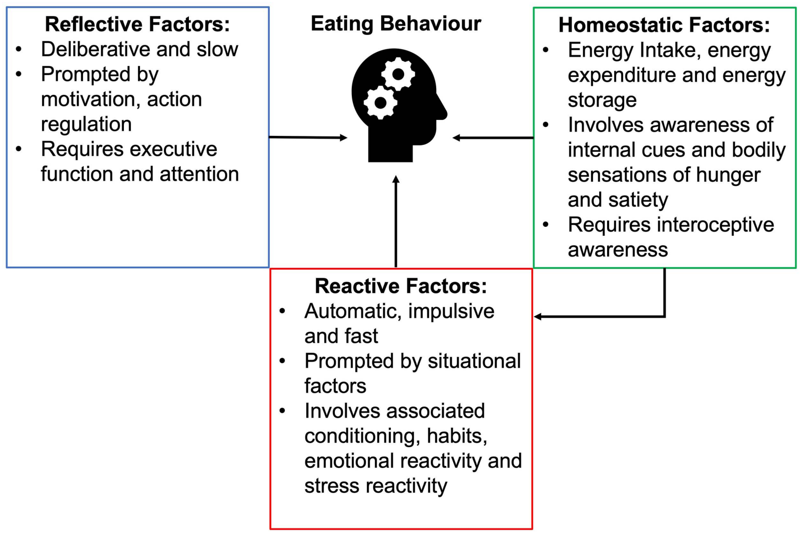
There is a disconnect between physiological and neurobiological approaches to eating behavior, and psychological theories are a critical issue that limits a more comprehensive understanding of eating behavior. The lack of integration between these approaches has led to fragmented knowledge and limited insights into the complex interplay between physiological processes and psychological factors that influence eating behavior. Consequently, there has been little attempt to unify these perspectives, further exacerbating the gaps in our understanding. Developing a framework that systematically identifies the domains of EBTs is a crucial next step toward bridging the gap between physiology, neurobiology, and psychology.

EBTs Framework

The last aim of this review is to develop a framework of EBTs that explains observed behaviors in the context of key psychological theories, physiology, and neurobiology of appetite. As previously described, many approaches do not highlight the importance of both psychology and physiology of eating behavior, and therefore, there is a need to move toward a more integrated approach. Current theories that explain EBTs appear to be made up of theories that can be considered as "fragments" of dual-process models. Many of

Figure 1

A Graphical Representation of an Integrated Physiological, Neurobiological, and Psychological Model of Energy Balance Regulation



Note. Directly related to eating behavior are reactive processes of learned and habitual behaviors that respond to environmental and psychological cues such as food availability and stress, respectively. Another domain represented in this diagram is reflective factors, which involve deliberative and cognitive modification of attitudes, beliefs, and intentions that aim to change eating behavior. This domain often has less influence on energy balance than other domains, and often these strategies conflict with reactive and/or homeostasis and hedonic factors, which also renders them less effective. Last, the diagram also recognizes that elements of the energy balance system influence both reflective eating behavior and reactive components of eating behavior, which are represented in the homeostatic domain. See the online article for the color version of this figure.

these theories regard eating behavior as comprising two processes involving automatic and deliberate behavior. Physiological theories, on the other hand, highlight the importance of biological signals influencing behavior but also fail to explain how other important nonphysiological factors, such as environmental cues and emotions, drive eating behavior. Furthermore, newer integrative frameworks, such as uncontrolled eating, are driven by data on eating behavior. However, they are not anchored onto an a priori theory of eating behavior.

The two major systems that influence motivation to eat are conscious processes (reflective, cognitive factors) and automatic processes (reactive factors). These two domains also interact with physiological signaling, which includes homeostatic feedback. Thus, EBTs can be grouped under three higher order, conceptual domains: reflective, reactive, and homeostatic processes (see Figure 2). These domains are labeled. However, other synonymous labels could be used, depending on the field of study. For example, cognitive psychologists may refer to reactive eating as automatic eating. While it is acknowledged that different labels could be used for these factors, it would be useful for future research to take a consultative approach with experts to discuss and clarify these labels.

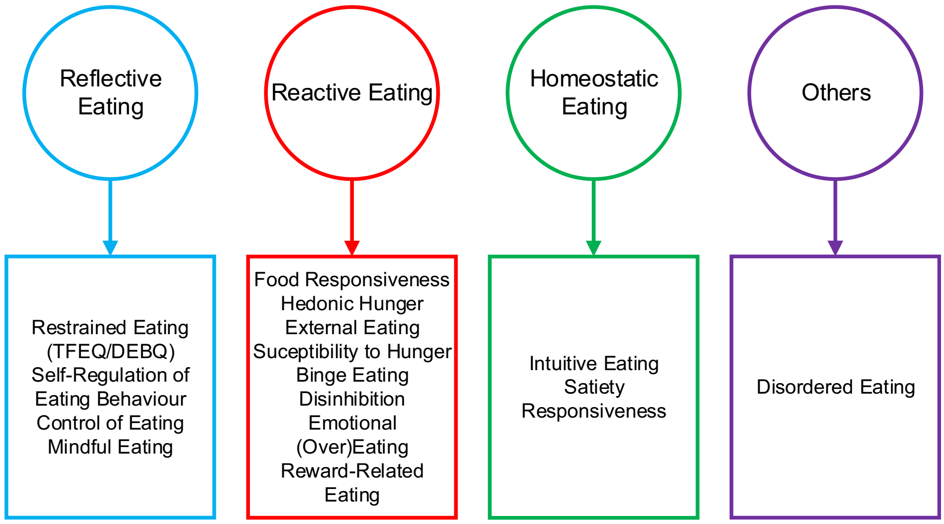
Reflective processes involve deliberative cognitive reasoning with some degree of analytical awareness of eating behavior. This process is governed by slow, logical, and sequential thinking and includes strategies of behavior change. EBTs influenced by reflective processes refer to conscious control over food intake, involving thought, decisions, or control. See Figure 2 for measures that could be influenced by reflective processes. The reflective domain has less influence on energy balance than other domains due to a combination of environmental, physiological, and hedonic factors that work to encourage overconsumption and thus a positive energy balance.

Many weight management interventions use behavior change techniques (BCTs) as a method of achieving weight loss. Several of these BCTs can be considered examples of reflective processes. For example, strategies to improve or develop self-efficacy, self-motivation, and self-monitoring involve developing competencies, strategies, and skills aimed at resolving challenging situations and reducing emotional stress and cognitive control to exhibit specific behaviors (e.g., self-weighing or restricting energy intake). Additionally, engaging in consistent exercise requires effort, motivation, and determination, which are all examples of reflective processes (J. Stubbs et al., 2011). Unfortunately, evidence shows that maximum weight loss through behavior change interventions is achieved at 6 months, and after this point, body weight increases gradually to baseline levels (Garcia Ulen et al., 2008). Research also shows that other domains work to oppose the reflective domain that comes from an understanding of the barriers to weight loss. Indeed, one study found that the most important perceived barriers to dieting for weight loss were food craving, stress, and depression (Sharifi et al., 2013).

Reactive processes involve automatic eating behavior, which is governed by external stimuli including emotions, impulses, habits,

Figure 2

A Conceptual Framework of EBTs



Note. A conceptual framework of EBTs that suggests EBTs can be grouped under three higher order conceptual domains: reactive, reflective, and homeostatic eating. This framework also understands that the labels given to these domains can be swapped for many other labels depending on the context for which this framework is being used. Last, there are some EBTs that this framework does not attempt to explain, and this is recognized in the figure. TFEQ = Three Factor Eating Questionnaire; EBTs = eating behaviour traits; DEBQ = Dutch Eating Behavior Questionnaire. See the online article for the color version of this figure.

and desires. Evidence suggests that a large amount of human behavior is reactive and learned, and the environment can generate cues, prompts, and stimuli that drive learned motivations to eat (Cohen & Babey, 2012). EBTs influenced by reactive processes refer to automatic influences over food intake that indicate the effect of emotions, desires, prompts, cues, impulses, and habits (see Figure 2 for examples). A variety of cues in the environment, for example, the sight of food, can become associated with eating, which provokes learned cue reactivity (Jansen et al., 2016). Studies have shown that food cue reactivity increases the risk of overeating, weight gain, and relapse after weight loss (Jansen, Havermans, & Nederkoorn, 2011; Jansen, Nederkoorn, et al., 2011).

Reactive processes also include eating behaviors that are driven by habits, which can explain why reactive eating behavior is pursued even in the absence of a food cue. Habits are behaviors that are frequently repeated over time and are learned sequences that have been reinforced by rewarding experiences, which are largely outside of conscious awareness because they have become subsumed to automaticity (Neal et al., 2006). van Koningsbruggen et al. (2013) found that tempting food cues increase effortful behavior toward high-calorie food among unsuccessful but not successful dieters. Whereas, tempting food cues increased efforts toward low-calorie food in successful dieters but did not affect unsuccessful dieters.

Additionally, when an individual engages in a weight loss attempt, there are physiological responses that are more pronounced in response to negative rather than positive energy balances. Some of these responses involve active compensation (e.g., decreased total daily energy expenditure, including reductions in RMR and nonexercise activity energy expenditure), and some involve physiological processes that influence behavior (e.g., decreased physical activity and increased energy intake; R. J. Stubbs & Turicchi, 2021). There are also reactive psychological and behavioral responses to weight loss, including increased motivation to eat and compensatory increases in energy intake and compensatory decreases in energy expenditure. For example, Polidori et al. (2016) found that appetite increased by ∼100 kcal/day above baseline per kg of lost weight. This demonstrates how difficult it is to successfully maintain weight loss due to increases in reactive processes such as increased appetite and suppression of energy expenditure that exist partly because of living in an obesogenic environment.

Importantly, the current approach using self-report measures of eating behavior is limited because reactive eating processes occur automatically. As such, the capability of individuals in being able to answer questions on their automatic eating behaviors can be questioned. For example, in the context of emotions and eating behavior, Winkielman et al. (2005) argued that "for an emotion to be unconscious, people must not be able to report their emotional reaction at the moment it is caused." This would suggest that self-report items are unable to access reactive, automatic eating behaviors. However, they also argued that there can be evidence of the emotional reaction in one's behavior, physiological responses, or the subjective impression after the event (Winkielman et al., 2005). Thus, items used to measure reactive EBTs measure thoughts, feelings, motivations, and behaviors, which can be used to measure automatic EBTs. Though it must be acknowledged that there are limitations to this approach, self-report measure of EBTs do allow us to access information about a person that is rich with their eating motivations and contains more information that other people may not be aware of (Paulhus & Vazire, 2007).

Homeostatic eating refers to the conscious awareness of internal cues and bodily sensations of hunger and satiety. It is defined as eating determined by energy deficits (Lowe & Butryn, 2007). This domain therefore reflects the physiological theories of homeostasis, which argue that eating behavior is driven by physiological signals that respond to a negative energy balance (Watts et al., 2022). No EBT has been specifically designed to measure homeostatic eating, nor indeed types of motivation to eat in response to energy imbalances, but there are scales that measure conscious awareness of sensations that can be considered related to or proxy measures of homeostatic eating (see Figure 2). Items on the reliance on hunger and satiety subscales, for example, measure whether an individual can trust and rely on their internal hunger and satiety cues, which are central features of homeostasis.

Homeostatic eating is distinct from reflective eating because it measures the ability to listen to and act on internal and biological hunger and satiety signals, which is not the same as cognitive control of eating. This may require an individual to have a level of interoceptive awareness to assess attention to internal signals. However, homeostatic eating may also require no interoceptive awareness if an individual is unconsciously compensating for their energy intake over a period of weeks and months (Polidori et al., 2016). An important theoretical consideration of the homeostatic eating domain is that overeating and obesity can be explained by the asymmetry of body weight regulation.

Overall, there is little evidence that weight gain produces active compensation of behavior to reduce energy intake. However, there is evidence that weight loss produces physiological and behavioral responses that unconsciously increase energy intake that escalate as weight loss escalates (R. J. Stubbs & Turicchi, 2021). The mechanisms involved differ depending on the level of adiposity of an individual. What is likely driving energy intake leading to weight gain are compensatory responses, responses to environmental cues related to hedonics, and habits. The evidence suggests that hunger does not lead to weight gain, but it does influence weight loss, such that the extent to which an individual feels hungry depends on their motivations to eat, including whether they feel deprived of their psychological needs (R. J. Stubbs & Turicchi, 2021). In the context of a negative energy balance, it is likely that reactive processes override homeostatic processes to promote increased energy intake, therefore inhibiting weight loss success. In the context of overeating leading to significant weight gain, there is little evidence that increased body weight or fat mass exerts homeostatic negative feedback to decrease energy intake, appetite, and hunger in order to regulate body weight (R. J. Stubbs et al., 2018). This indicates that in individuals with a higher BMI, homeostatic processes fail to compensate for a positive energy balance, which can lead to overweight and obesity.

Discussion

This conceptual article aimed to begin developing a comprehensive framework to identify the core, latent constructs underlying human EBTs by drawing from key psychological, physiological, and neurobiological theories and models. The article began by summarizing and evaluating influential theories of eating behavior and EBTs that have been developed from theory. This review has highlighted a lack of an agreed framework or standardized set of measures for describing motivations for eating. Additionally, the current state of EBTs has some limitations. The abundance of EBTs has led to a lack of clarity and conceptual overlap, making it difficult to understand the underlying mechanisms of EBTs. Further, this review has drawn attention to the importance of researchers understanding what exactly a scale is measuring and elucidating why a certain scale is being used to measure a specific outcome. A main outcome of the Psychological Accounts of Eating Behavior section of this review suggests that many theories draw on concepts from dual-process models, which propose that eating behavior is driven by two competing processes. One process involves cognitive, reflective, and decision-making abilities, which include sensitivity and responsivity to internal cues. The other process involves automatic, reactive, and impulsive behavior, which include reacting to external cues in the environment, eating for pleasure/comfort, and overeating to reduce negative or enhance positive emotions. Importantly, there are considerable overlaps between theories of eating behavior, and many newer theories incorporate concepts from older theories.

It is hoped that EBTs may be better understood using this framework. This is because the framework facilitates insight as to which EBTs overlap and which are distinct. However, it is also highlighted that there are other areas of appetite control and eating behavior that are not explained by this framework because they are beyond the scope of this article. For example, eating disorders are not included within this framework of EBT. Some EBTs have been developed to measure abnormal eating behaviors. The Yale Food Addiction Scale (Gearhardt et al., 2009), for example, adapts diagnostic criteria from the *Diagnostic and Statistical Manual of Mental Disorders* to measure the construct of food addiction. Disordered patterns of eating behavior do not occur in normal eating behavior, and in the case of food addiction, there is some evidence to suggest food and drugs of abuse utilize similar pathways in the brain: opiate and dopamine systems (Hoebel et al., 1999; Nieto et al., 2002). Additionally, the diagnostic criteria for substance abuse are similar to the diagnostic criteria for binge eating disorder (Gold et al., 2003).

There is an ongoing debate about the concept of food addiction, whether highly processed foods are addictive, and whether this can help our understanding of overeating and obesity. Evidence for the concept of food addiction includes similarities between highly processed foods and drugs of abuse, such as that both are highly reinforcing and some (but not all) individuals consume them compulsively (Gearhardt & Hebebrand, 2021b). Additionally, highly processed foods are associated with craving, reduced control over consumption, and continued use despite negative consequences, which are all behavioral indicators of addiction (Gearhardt & Hebebrand, 2021b). However, there is also debate against the concept of food addiction. For example, treatment for somatic and mental disorders requires exclusion, which is not possible because humans have a physiological need to ingest sufficient calories to maintain body weight (Hebebrand & Gearhardt, 2021). The concept of food addiction also lacks validation and has not led to any new and successful treatments for obesity (Hebebrand & Gearhardt, 2021). Overall, researchers do agree that addictive-like eating exists; however, there are disagreements about whether highly processed foods are addictive and the societal implications for the food addiction concept (Gearhardt & Hebebrand, 2021a). Due to the ongoing debate about food addiction and because disordered eating goes beyond the scope of the current article, disordered eating is subsumed into the periphery of the conceptual diagram of EBTs (Figure 2), which includes examples of "aberrant" eating behavior. Disordered eating therefore appears more of an outcome domain than a causal/explanatory domain, and for these reasons, EBTs that measure disordered eating are not currently included within the framework of EBTs.

One notable framework proposed by Vainik et al. (2015) suggests that many EBTs could be capturing the same underlying construct, termed "uncontrolled eating," which involves increased appetite and decreased self-control. The authors developed a continuum model of uncontrolled eating, incorporating five EBTs: eating impulsivity, hedonic hunger, emotional eating, disinhibition, and binge eating, which measure different ranges of values along this continuum. This framework has begun to develop a comprehensive perspective of the current literature by suggesting that EBTs are measuring different degrees of lack of control of eating. However, future research is required to understand if this framework can account for a wider variation of EBTs. While there have been some attempts to integrate EBTs into a conceptual framework, much of this research is data-driven, meaning it lacks theoretical underpinnings.

The next sections (Physiology of Eating Behavior and Neurobiology of Eating Behavior) of this review summarized key physiological and neurobiological concepts with the aim of incorporating these concepts with psychological theory. The physiological perspective of appetite control and energy balance reveals the importance of homeostatic processes, manifesting as physiological hunger and satiety responses, in understanding eating behavior. These processes constitute essential traits that should be included in a framework of eating behavior. Some existing EBTs already measure physiological hunger, such as the hunger subscale of the AEBQ. Additionally, EBTs like the satiety responsiveness scale from the AEBQ capture an individual's sensitivity to internal levels of satiety, representing homeostatic processes. Neurobiological studies further emphasize that appetite regulation involves not only homeostatic mechanisms but also hedonic and reward-based pathways. Pleasure and reward play a central role in controlling or lack of control over energy intake. Neurobiological mechanisms, including food reward, executive function, affect-related eating, and stress-related eating, should also be incorporated into EBTs to provide a comprehensive understanding of eating behavior. Emotional eating, which has been studied extensively, has led to the development of EBTs that measure various aspects of emotional eating.

A critical issue, which is emphasized by this review, lies in the lack of integration between physiological and neurobiological approaches to eating behavior and psychological theories. This disconnect hinders a comprehensive understanding of the complex interplay between physiological processes and psychological factors that influence eating behavior. To address these limitations, this article proposed a new framework that categorizes EBTs into three higher order domains: reflective processes, reactive processes, and homeostatic eating. These domains are labeled. However, several other synonymous labels could also be used, depending on the context in which this framework is used.

Reflective processes encompass conscious cognitive reasoning and behavior change strategies, but their influence on energy balance is limited due to environmental, physiological, and hedonic factors that encourage overconsumption. Weight loss interventions often rely on reflective strategies, but evidence shows that maintaining weight loss is challenging due to increased appetite/motivation to eat and reduced energy expenditure, which can lead to weight gain and regain. Reactive processes involve automatic eating behavior triggered by external stimuli, emotions, habits, and desires. These processes can override homeostatic signals, also leading to overeating and weight gain. Environmental cues, learned motivations, and habits play a significant role in reactive eating behavior, making it difficult to resist overeating. Last, homeostatic eating refers to the conscious awareness of internal hunger and satiety cues and is a domain less explored by existing EBTs. Homeostatic eating relies on physiological signals responding to energy deficits, but EBTs targeting this domain specifically do not exist. This therefore provides an avenue for further study. Future research should aim to develop EBTs that specifically measure awareness of internal hunger and satiety cues. It is noted that homeostatic eating may or may not require interoceptive awareness, depending on whether an individual is unconsciously compensating for their energy intake over the long term. Additionally, the influence of homeostatic processes likely also depends on whether an individual is in a positive or negative energy balance and the strength of an energy imbalance. Consequently, unpicking the complexity of homeostatic processes could provide valuable insights into energy balance regulation and obesity.

Conclusions and Future Directions

To conclude, this article proposes a starting point to developing a comprehensive framework of EBTs that encompasses reflective, reactive, and homeostatic processes. This framework has helped to understand EBTs better by identifying the core, latent constructs underlying EBT measurements as well as aligning EBT constructs with existing theory. By considering both psychological and physiological dimensions, this framework aims to bridge existing gaps in our understanding of human eating behavior.

It is important to highlight that the current review is a starting point to a comprehensive framework of EBTs and not a final framework. Thus, the planned next steps for the work are to take a more inclusive and consultative approach to build on the framework with input from other experts in the field. For future research, we aim to use the Delphi method and construct a survey using the current framework to generate a first round of feedback to try and capture elements that may be missing or underdeveloped. We hope to use this provisional model as a starting point for discussion, with the aim of achieving consensus about the framework and identifying any constructs that require further development.

A similar and comprehensive approach to the Delphi method has been used in the behavior change field with the recent development of behavior change ontology. It has been identified that descriptions of behavior change interventions vary widely, and this hinders the ability to compare or replicate studies and develop and evaluate the effectiveness of interventions (Marques et al., 2023). To address this, Michie et al. (2013) developed the behavior change taxonomy (BCTTv1). BCTs are the smallest elements of a behavior change intervention, and when developing the BCTTv1, a Delphi-like method was used to include the input of 400 experts, which led to 93 BCTs being organized into 16 higher order groupings. Since its development, the BCTTv1 has been widely cited in more than 5,000 published studies, and it has enabled a structured method for designing and evaluating behavior change interventions. The label of "v1" highlights that the taxonomy would need development as experts and users gave feedback and as the field advanced. This user feedback included responses from surveys, researchers, and interview-based consultations, which improved the labels and definitions and developed the structure of the taxonomy (Corker et al., 2022).

Another necessary step to improve the BCTTv1 was the development of a behavior change technique ontology (BCTO). Ontologies are structures for expressing knowledge by defining entities and their relationships (Arp et al., 2015). They offer a more comprehensive way of representing information than taxonomies and are designed to be added to with new information. The development of the BCTO was also informed by user feedback, which suggested the need for additional BCTs, amendments to labels, definitions, and groupings, and improvements to improve clarity (Marques et al., 2023). Taking an ontology approach to behavior change has extended and improved upon the BCTTv1, and the authors note that the BCTO should still be updated and improved using an ongoing and collaborative process. Although the field of EBTs is much smaller than that of the behavior change field, using similar methods to the BCTO would also benefit the EBT field and improve the current framework in this article.

Indeed, a Delphi-like approach has also recently been used in the field of personality. Irwing et al. (2023) have developed a comprehensive taxonomy of unique personality facets, including 1,772 personality items taken from multiple personality inventories. The inventory, termed the Facet-level Multidimensional Assessment of Personality, Version 1, was developed through an iterative, multistage, multimethod approach. After identifying 136 potential factors for inclusion within the taxonomy, the authors conducted a panel review whereby each panel member individually evaluated each pair of facets until 10 facet pairs in a row were thought to be unique. The panel members also identified problems with the conceptual coherences of the included scales and systematically compared the item content of each facet with all other facets. Overall, taking a consultative approach, such as including a panel review, has been found to be an effective procedure (DeMaio & Landreth, 2004). Accordingly, taking inspiration from the personality field and behavior change, approaches like the Delphi method are needed to advance future research.

Further research and development of new EBT measures within this framework will also enable researchers to gain deeper insights into the underlying motivations for eating and contribute to the development of effective interventions for managing eating behavior and obesity. Further research is also required to begin refining the number of EBTs used in studies so that only the most internally and externally valid EBTs are used. An important aim is for EBTs to have better explanatory power in human eating behavior and obesity research. Future research should also test whether this framework is empirically supported through using data analysis techniques such as factor analysis. This will enable validation of the conceptual framework by confirming whether the hypothesized relationships and patterns exist in data sets that measure EBTs in real people. It is also important to note that while the majority of this article focuses on EBTs that measure overeating, which is important for the study of obesity, it is also important to consider the implications of this framework beyond the field of obesity. Therefore, future research should consider how the current article and the proposed framework can be used to explain and improve diet choice, health and well-being, food insecurity, and healthy eating in childhood development. Overall, these next steps are vital to further understand how human eating behavior is influenced by different domains/latent constructs and how these domains interact with each other.

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